90° Pipe Brace Bracket: Static FEA Report

Part: 90degree_12in_0.133_Pipe_Brace_Bracket_V1 (Default<As Machined>) | Study: Static 1 | Date: November 3, 2026

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1. Summary

A 1 in NPS Schedule 40 steel pipe bent 90° on a 12 in centerline radius connects two 4 × 4 × 3/8 in bolted mounting plates. The bracket is intended as a rigid metal support (brace) that carries a load from an attached component back into a floor, wall or frame. A 1,000 N (225 lbf) load was applied to the free end plate with the base plate restrained at its bolt holes and bearing face.

Result	Value	Comment
Peak von Mises stress	375.9 MPa (54.5 ksi)	Localized at pipe-to-plate junction; exceeds yield
Nominal bend stress (FEA field)	≈110–190 MPa	Lower bend near base plate
Nominal bend stress (hand calc)	140 MPa (20.3 ksi)	M = F·R at base, S = 0.1328 in³
Max resultant displacement	4.26 mm (0.168 in)	At free end plate
Max equivalent strain	1.09×10^{-3}	Element 4311
Factor of safety (yield)	0.59 peak / 1.2–2.0 nominal	Yield 220.6 MPa
Total mass	2.54 kg (5.61 lb)	Two bodies, plain carbon steel

2. Geometry

Feature	Dimension
Pipe	1 in NPS Sch 40: OD 1.315 in, ID 1.049 in, wall 0.133 in
Bend	90°, 12.0 in centerline radius (≈18.8 in arc length)
Mounting plates (2)	4.00 × 4.00 × 0.375 in, 0.10 in corner radius
Bolt holes	Ø0.531 in (17/32 in) thru, 4 per plate (8 total), 2.50 in square pattern on plate center; clearance for 1/2 in bolts
Pipe bore in end plate	Ø1.330 in thru, centered (0.0075 in radial clearance on 1.315 in OD)
Bodies	Body 1: pipe + base plate, 1.891 kg, 2.424×10^{-4} m³. Body 2: end plate, 0.653 kg, 8.374×10^{-5} m³



Figure 1. Model with fixtures (green, base plate) and load (magenta, end plate).

3. Setup

Material: Plain Carbon Steel (linear elastic isotropic)

E = 210 GPa, v = 0.28, G = 79 GPa, ρ = 7,800 kg/m³, Sy = 220.6 MPa (32.0 ksi), Su = 399.8 MPa (58.0 ksi), α = 1.3 × 10^{-5} /K. Comparable to ASTM A53 Gr. B pipe (Sy ≈ 240 MPa) and A36 plate (Sy ≈ 250 MPa), so results are slightly conservative for those grades.

Loads and fixtures

Item	Entity	Type	Reaction (N)
Fixed-1	4 bolt-hole bores, base plate	Fixed geometry	0.71
Roller/Slider-1	Underside of base plate	Roller/slider	997.3
Force-1	Face of end plate	1,000 N along ref. edge (vertical)	n/a

Sum of reactions = 999.999 N against 1,000 N applied; free-body residual 0.0021 N. Equilibrium is satisfied. Nearly all load is carried through bearing on the base plate face, which is the expected path for a bolted base.

Interaction and mesh

Global contact: bonded, independent mesh (end plate bonded to pipe). Blended curvature-based solid mesh, high quality (16 Jacobian points): 41,720 nodes, 23,845 elements, element size 0.077–0.589 in. 93.5% of elements have aspect ratio < 3, 0.36% exceed 10 (max 430), 0% distorted. Mesh time 4 s. Large displacement, friction, soft spring and inertial relief off; thermal effects on at 298 K zero-strain reference with no temperature load applied.

4. Assumptions

1. Linear elastic, small-displacement, static response; no plasticity, buckling, fatigue or dynamic amplification.
2. Material is homogeneous, isotropic plain carbon steel at room temperature (298 K); no thermal gradient, so thermal stress is zero.
3. Welds are idealized: the pipe-to-base joint is modeled as continuous material and the end plate as fully bonded. No weld fillets, heat-affected zone or residual stress are modeled.
4. Bolts are not modeled. Fixing the hole bores and adding a roller on the bearing face approximates preloaded 1/2 in bolts on a rigid foundation; real foundation and bolt stretch add compliance.
5. The 1,000 N load is uniformly distributed over the end-plate face and acts vertically. Self-weight (24.9 N) is neglected. Results scale linearly with load and apply equally to a downward load of the same magnitude.
6. Nominal pipe dimensions; no corrosion allowance, mill tolerance or bend-induced wall thinning or ovality.

5. Results

The stress plot shows a smooth bending field along the bend, rising toward the base where the moment arm is largest (≈12 in). The 375.9 MPa peak (node 18098) is a small, localized hot spot at the sharp, un-filleted pipe-to-plate corner. This is a geometric stress concentration that is amplified by mesh refinement (a singularity), so it should not be read as the true nominal stress. It does flag the weld toe as the critical location.

Hand check: M = 1,000 N × 0.3048 m = 304.8 N·m. S = π(1.315⁴ − 1.049⁴)/(32 × 1.315) = 0.1328 in³ = 2.18 × 10^{-6} m³. σ = M/S = 140 MPa, inside the 110–190 MPa FEA field at the bend. Tip deflection from bending only, quarter-circle cantilever: δ = πFR³/(4EI) with I = 0.0873 in⁴ gives 2.9 mm. FEA gives 4.26 mm; the difference comes from base-plate bending between the bolts, rotation of the end plate and joint compliance.

Bolts: assuming the base plate pivots about its compression edge, the two tension bolts each carry ≈1.85 kN (415 lbf), far below the proof load of a 1/2-13 Grade 5 bolt (≈53 kN).



Figure 2. von Mises stress (deformation scale 11.1×). Peak 3.759 × 10⁸ N/m² at the pipe-plate junction.

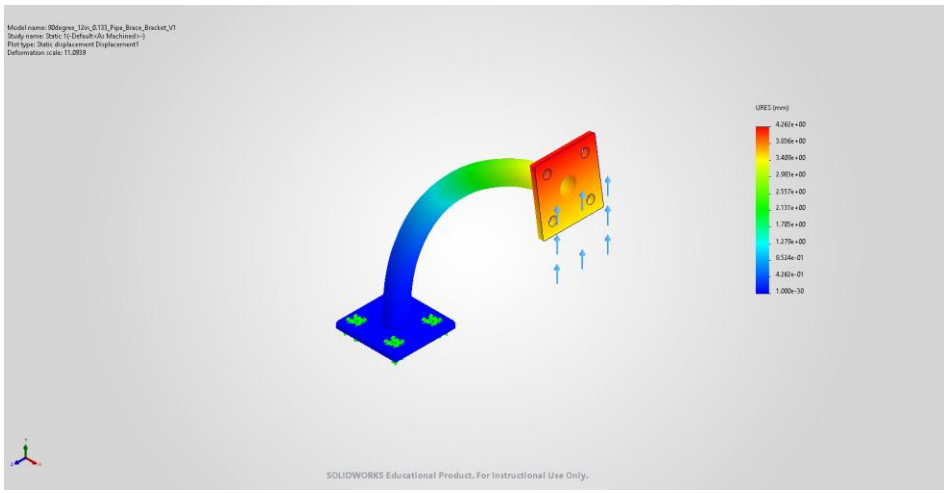


Figure 3. Resultant displacement. Maximum 4.262 mm at the free end plate.

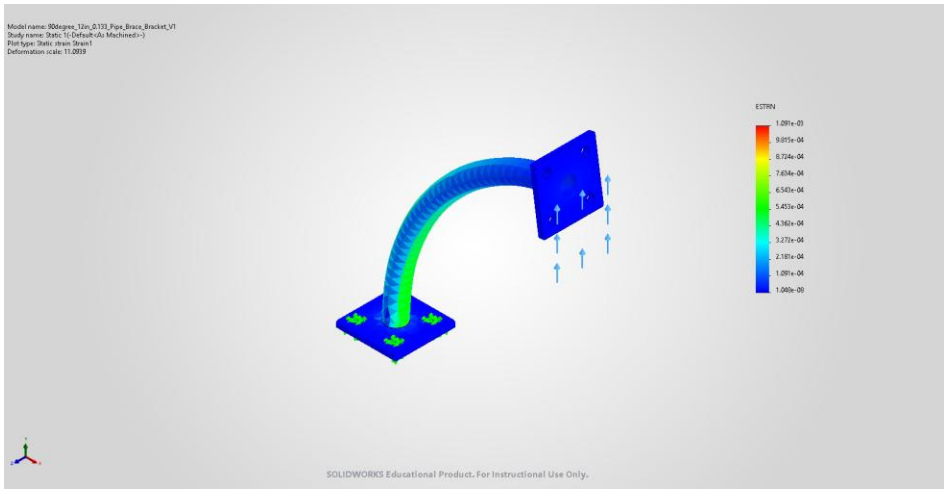


Figure 4. Equivalent strain. Maximum 1.091 × 10⁻³.

6. Conclusions

7. **Pipe body is adequate at 1,000 N, but with thin margin.** Nominal bending stress of 140 MPa gives FOS ≈ 1.6 on yield (1.2 at the upper end of the FEA field). This is below the 1.67 typically used for steel members in allowable-stress design.

8. **The pipe-to-plate weld is the critical location.** The 376 MPa local peak (FOS 0.59) means local yielding at the weld toe is likely under this load in the as-modeled sharp-corner geometry. Under repeated loads such as vibration or pipe surge, this spot would be where fatigue cracks start.
1. **Stiffness is acceptable for a brace.** 4.3 mm (0.17 in) tip deflection over an approximately 12 in reach is about L/72: fine for a static restraint, too flexible for alignment-critical mounting.
2. **Plates and fasteners are not limiting.** 3/8 in plate and four 1/2 in bolts per plate have large margin; the bend governs.
3. **Recommendation.** Rate the current design at ≤ 700 N (155 lbf) working load for FOS ≥ 1.67, or keep 1,000 N and either use 1 in Sch 80 pipe ($\sigma \approx 116$ MPa, FOS 1.9) or 1-1/4 in Sch 40 ($\sigma \approx 79$ MPa, FOS 2.8). Specify a full-circumference fillet weld (≥ 3/16 in) or a gusset at the base. Model the weld fillet and run a mesh convergence study at the junction before final release.